

Column Space, Null Space and Rank-Nullity

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Column space and null space

Definition 0.1 (Column space (range, image)). The **column space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\tilde{x} : \tilde{x} \in \mathbb{R}^n\},$$

the set of all vectors $\mathbf{A}\tilde{x}$ in $\mathbb{R}^m$. It is also called the **range** or **image** of $\mathbf{A}$. The **row space** of $\mathbf{A}$ is $\mathcal{C}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^n$.

Example 0.1 (The column space of a rank-one matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any $\tilde{x} \in \mathbb{R}^3$,

$$\begin{aligned} \mathbf{A}\tilde{x} &= (x_1 - 2x_2 - 2x_3, 3(x_1 - 2x_2 - 2x_3)) \\ &= (x_1 - 2x_2 - 2x_3)(1, 3) \end{aligned}$$

([Definition 11 in Matrices](#)). The number $x_1 - 2x_2 - 2x_3$ can be any real number (take $x_2 = 0$ and $x_3 = 0$), so $\mathcal{C}(\mathbf{A}) = \{c(1, 3) : c \in \mathbb{R}\}$, a line in $\mathbb{R}^2$. The vector $(1, 0)$ is not in $\mathcal{C}(\mathbf{A})$: $c(1, 3)$ has second entry $3c$, which is 0 only if $c = 0$, and then the first entry is 0, not 1.

Theorem 0.1 (The column space is the span of the columns). *If $\mathbf{A}$ is an $m \times n$ matrix with columns $\tilde{a}_1, \dots, \tilde{a}_n$, then*

$$\mathcal{C}(\mathbf{A}) = \text{span}\{\tilde{a}_1, \dots, \tilde{a}_n\},$$

so $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$, and the row space $\mathcal{C}(\mathbf{A}^\top)$ is a subspace of $\mathbb{R}^n$.

i Proof

Proof. By [Theorem 3 in Subspaces and Rank](#), $\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \cdots + x_n\tilde{a}_n$, and as $\tilde{x}$ ranges over $\mathbb{R}^n$, the coefficients $x_1, \dots, x_n$ range over all lists of n numbers. So the set of all $\mathbf{A}\tilde{x}$ ([Definition 0.1](#)) is the set of all linear combinations of the columns, which is their span ([Definition 5 in Subspaces and Rank](#)). A span is a subspace ([Theorem 2 in Subspaces and Rank](#)). Applying the same argument to $\mathbf{A}^\top$, whose columns are the rows of $\mathbf{A}$, shows the row space is a subspace of $\mathbb{R}^n$. $\square$

Example 0.2 (Reading off the column space and row space from the columns and rows). For $\mathbf{A}$ in [Example 0.1](#), the columns are $(1, 3)$, $(-2, -6) = -2(1, 3)$ and $(-2, -6) = -2(1, 3)$, so by [Theorem 0.1](#) $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$, the same line found in [Example 0.1](#). The rows are $(1, -2, -2)$ and $(3, -6, -6) = 3(1, -2, -2)$, so the row space is $\text{span}\{(1, -2, -2)\}$, a line in $\mathbb{R}^3$.

Definition 0.2 (Null space (kernel)). The **null space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^n : \mathbf{A}\tilde{x} = \tilde{0}_m\},$$

the set of [solutions](#) of the linear system $\mathbf{A}\tilde{x} = \tilde{0}_m$. It is also called the **kernel** of $\mathbf{A}$. The **left null space** of $\mathbf{A}$ is $\mathcal{N}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^m$.

Example 0.3 (The null space and left null space of a rank-one matrix). For $\mathbf{A}$ in [Example 0.1](#), $\mathbf{A}\tilde{x} = (x_1 - 2x_2 - 2x_3, 3(x_1 - 2x_2 - 2x_3))$, so $\mathbf{A}\tilde{x} = \tilde{0}_2$ exactly when $x_1 = 2x_2 + 2x_3$. Writing

$$\begin{aligned}\tilde{x} &= (2x_2 + 2x_3, x_2, x_3) \\ &= x_2(2, 1, 0) + x_3(2, 0, 1)\end{aligned}$$

shows $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$, a plane in $\mathbb{R}^3$.

For the left null space, $\mathbf{A}^\top \tilde{y} = (y_1 + 3y_2, -2(y_1 + 3y_2), -2(y_1 + 3y_2))$, which is $\tilde{0}_3$ exactly when $y_1 = -3y_2$, so $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$, a line in $\mathbb{R}^2$.

Theorem 0.2 (A null space is a subspace). *For any $m \times n$ matrix $\mathbf{A}$, $\mathcal{N}(\mathbf{A})$ ([Definition 0.2](#)) is a subspace of $\mathbb{R}^n$ ([Definition 4 in Subspaces and Rank](#)).*

i Proof

Proof. $\mathbf{A}\tilde{0}_n = \tilde{0}_m$, so $\tilde{0}_n \in \mathcal{N}(\mathbf{A})$, and the null space is not empty. If $\tilde{u}, \tilde{w} \in \mathcal{N}(\mathbf{A})$ and $c \in \mathbb{R}$, then

$$\begin{aligned}
\mathbf{A}(\tilde{u} + \tilde{w}) &= \mathbf{A}\tilde{u} + \mathbf{A}\tilde{w} && \text{(Theorem 6 in Matrices)} \\
&= \tilde{0}_m + \tilde{0}_m && (\tilde{u} \text{ and } \tilde{w} \text{ are in the null space}) \\
&= \tilde{0}_m && (\text{adding } \tilde{0}_m \text{ changes nothing})
\end{aligned}$$

and, for each entry $i = 1, \dots, m$,

$$\begin{aligned}
(\mathbf{A}(c\tilde{u}))_i &= a_{i1}(cu_1) + \dots + a_{in}(cu_n) && \text{(Definition 11 in Matrices)} \\
&= c(a_{i1}u_1 + \dots + a_{in}u_n) && (\text{factor } c \text{ out of each term}) \\
&= c(\mathbf{A}\tilde{u})_i && \text{(Definition 11 in Matrices)} \\
&= c \cdot 0 && (\tilde{u} \text{ is in the null space}) \\
&= 0, && (\text{arithmetic})
\end{aligned}$$

so $\mathbf{A}(c\tilde{u}) = \tilde{0}_m$. Therefore $\tilde{u} + \tilde{w}$ and $c\tilde{u}$ are in $\mathcal{N}(\mathbf{A})$. $\square$

Example 0.4 (The solutions of $\mathbf{A}\tilde{x} = \tilde{b}$ with $\tilde{b} \neq \tilde{0}$ are not a subspace). In Example 0.3, $(2, 1, 0)$ and $(2, 0, 1)$ are in $\mathcal{N}(\mathbf{A})$, and so is their sum $(4, 1, 1)$, since $4 - 2 \cdot 1 - 2 \cdot 1 = 0$. By contrast, the solutions of $\mathbf{A}\tilde{x} = (1, 3)$ for the same $\mathbf{A}$ are the vectors with $x_1 - 2x_2 - 2x_3 = 1$. That set does not contain $\tilde{0}_3$, so by Theorem 1 in Subspaces and Rank it is not a subspace.

Theorem 0.3 (The row space and null space share only the zero vector). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}.$$

i Proof

Proof. Both sets are subspaces, by Theorem 0.1 and Theorem 0.2, so both contain $\tilde{0}_n$ (Theorem 1 in Subspaces and Rank).

Now take any $\tilde{x}$ in both sets. Being in the row space means $\tilde{x} = \mathbf{A}^\top \tilde{u}$ for some $\tilde{u} \in \mathbb{R}^m$ (Definition 0.1), and being in the null space means $\mathbf{A}\tilde{x} = \tilde{0}_m$ (Definition 0.2). Then

$$\begin{aligned}
\tilde{x}^\top \tilde{x} &= (\mathbf{A}^\top \tilde{u})^\top \tilde{x} && (\text{substitute } \tilde{x} = \mathbf{A}^\top \tilde{u} \text{ in the first factor}) \\
&= (\tilde{u}^\top (\mathbf{A}^\top)^\top) \tilde{x} && \text{(Theorem 16 in Matrices)} \\
&= (\tilde{u}^\top \mathbf{A}) \tilde{x} && (\text{transposing twice returns the original matrix}) \\
&= \tilde{u}^\top (\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\
&= \tilde{u}^\top \tilde{0}_m && (\text{substitute } \mathbf{A}\tilde{x} = \tilde{0}_m) \\
&= 0. && (\text{every term of the inner product is } 0)
\end{aligned}$$

But $\tilde{x}^\top \tilde{x} = x_1^2 + \cdots + x_n^2$, which is 0 only when every $x_i = 0$. So $\tilde{x} = \tilde{0}_n$. □

Example 0.5 (The row space and null space in Example 0.3). For $\mathbf{A}$ in Example 0.1, the row space is $\text{span}\{(1, -2, -2)\}$ and the null space is $\text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). A vector $c(1, -2, -2)$ of the row space is in the null space only if

$$\begin{aligned} \mathbf{A}(c(1, -2, -2)) &= c(1 - 2 \cdot (-2) - 2 \cdot (-2), 3(1 - 2 \cdot (-2) - 2 \cdot (-2))) \quad (\text{Example 0.3's formula for } \mathbf{A}\tilde{x}) \\ &= c(9, 27) \quad (\text{arithmetic}) \end{aligned}$$

is $\tilde{0}_2$, that is, only if $c = 0$. So the two subspaces share only $\tilde{0}_3$.

Definition 0.3 (Gram matrix). The **Gram matrix** of an $m \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\mathbf{A}^\top \mathbf{A}$.

Example 0.6 (A Gram matrix and its entries). Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, with columns $\tilde{a}_1 = (1, 1, 0)$ and $\tilde{a}_2 = (0, 1, 1)$. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} && (\text{Definition 3 in Matrices}) \\ &= \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix} && (\text{Definition 7 in Matrices}) \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && (\text{multiply and add}) \end{aligned}$$

Row i of $\mathbf{A}^\top$ is column i of $\mathbf{A}$, so entry (i, j) of the Gram matrix is the dot product $\tilde{a}_i \cdot \tilde{a}_j$: for instance,

$$\begin{aligned} \tilde{a}_1 \cdot \tilde{a}_2 &= 0 + 1 + 0 \\ &= 1 \end{aligned}$$

is entry $(1, 2)$.

Theorem 0.4 ($\mathbf{A}^\top \mathbf{A}$ has the same null space as $\mathbf{A}$). For any $m \times n$ matrix $\mathbf{A}$,

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

Proof

Proof. $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$. If $\mathbf{A}\tilde{x} = \tilde{0}_m$, then

$$\begin{aligned}\mathbf{A}^\top \mathbf{A}\tilde{x} &= \mathbf{A}^\top \tilde{0}_m \\ &= \tilde{0}_n.\end{aligned}$$

$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$. If $\mathbf{A}^\top \mathbf{A}\tilde{x} = \tilde{0}_n$, then

$$\begin{aligned}0 &= \tilde{x}^\top \tilde{0}_n && \text{(every term of the inner product is 0)} \\ &= \tilde{x}^\top (\mathbf{A}^\top \mathbf{A}\tilde{x}) && \text{(substitute } \tilde{0}_n = \mathbf{A}^\top \mathbf{A}\tilde{x}) \\ &= (\tilde{x}^\top \mathbf{A}^\top) (\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\ &= (\mathbf{A}\tilde{x})^\top (\mathbf{A}\tilde{x}) && \text{(Theorem 16 in Matrices)} \\ &= (\mathbf{A}\tilde{x}) \cdot (\mathbf{A}\tilde{x}) && \text{(Example 6 in Matrices)} \\ &= \|\mathbf{A}\tilde{x}\|^2. && \text{(square both sides of Equation 2 in Vectors)}\end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so $\mathbf{A}\tilde{x} = \tilde{0}_m$. □

Example 0.7 (The null space of a Gram matrix). For $\mathbf{A}$ in Example 0.1,

$$\begin{aligned}\mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.\end{aligned}$$

Every row is a multiple of $(1, -2, -2)$, so $\mathbf{A}^\top \mathbf{A}\tilde{x} = \tilde{0}_3$ exactly when $x_1 - 2x_2 - 2x_3 = 0$, the same condition as for $\mathcal{N}(\mathbf{A})$ in Example 0.3.

Rank-nullity and rank factorization

Source

This section is adapted from Zhou (2024), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)).

Theorem 0.5 (The rank is the dimension of the column space). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\text{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

i Proof

Proof. Let $r = \text{rank}(\mathbf{A})$ (Definition 2 in Subspaces and Rank), and choose r linearly independent columns of $\mathbf{A}$; by Definition 2 in Subspaces and Rank, no larger set of columns is linearly independent. We show the chosen columns are a basis of $\mathcal{C}(\mathbf{A})$ (Definition 6 in Subspaces and Rank).

Every column is in the span of the chosen columns. A chosen column is in that span trivially. If some unchosen column $\tilde{a}_j$ were not in the span, then adding it to the chosen columns would give $r + 1$ linearly independent columns (Theorem 8 in Subspaces and Rank), more than r . So every column of $\mathbf{A}$ is in the span of the chosen columns.

The chosen columns span $\mathcal{C}(\mathbf{A})$. The span of the chosen columns is a subspace (Theorem 2 in Subspaces and Rank), so it is closed under addition and scalar multiplication, and it contains every linear combination of the columns of $\mathbf{A}$. By Theorem 0.1, those combinations make up $\mathcal{C}(\mathbf{A})$, so $\mathcal{C}(\mathbf{A})$ is contained in the span of the chosen columns. The chosen columns are themselves columns of $\mathbf{A}$, so their span is contained in $\mathcal{C}(\mathbf{A})$, and the two sets are equal.

The chosen columns are linearly independent and span $\mathcal{C}(\mathbf{A})$, so they are a basis of it, and $\dim(\mathcal{C}(\mathbf{A})) = r$ (Definition 8 in Subspaces and Rank). If $r = 0$, every column is $\tilde{0}_m$, $\mathcal{C}(\mathbf{A}) = \{\tilde{0}_m\}$, and both sides are 0. $\square$

Example 0.8 (The rank of the matrix in Example 0.1). For $\mathbf{A}$ in Example 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ (Example 0.2), and $(1, 3)$ is linearly independent because it is nonzero, so it is a basis and $\mathcal{C}(\mathbf{A})$ has dimension 1. So $\text{rank}(\mathbf{A}) = 1$, by Theorem 0.5. Directly: the first column $(1, 3)$ is linearly independent on its own, and any two columns are dependent, because each column is a multiple of $(1, 3)$.

Definition 0.4 (Nullity). The **nullity** of a matrix $\mathbf{A}$ is the dimension of its null space (Definition 0.2):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

Example 0.9 (The nullity of the matrix in Example 0.1). For $\mathbf{A}$ in Example 0.1, $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). Those two vectors are linearly independent: $c_1(2, 1, 0) + c_2(2, 0, 1) = (2c_1 + 2c_2, c_1, c_2)$, which is $\tilde{0}_3$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0. \end{aligned}$$

So they are a basis of $\mathcal{N}(\mathbf{A})$, and $\text{nullity}(\mathbf{A}) = 2$. The 2×2 identity matrix has nullity 0, because $\mathbf{I}\tilde{x} = \tilde{x}$ is $\tilde{0}_2$ only when $\tilde{x} = \tilde{0}_2$.

Theorem 0.6 (Rank-nullity theorem). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

i Proof

Proof. Let $\nu = \text{nullity}(\mathbf{A})$. Choose a basis $\tilde{x}_1, \dots, \tilde{x}_\nu$ of $\mathcal{N}(\mathbf{A})$, and extend it to a basis $\tilde{x}_1, \dots, \tilde{x}_\nu, \tilde{y}_1, \dots, \tilde{y}_{n-\nu}$ of $\mathbb{R}^n$ ([Theorem 9 in Subspaces and Rank](#)); it has n vectors, because every basis of $\mathbb{R}^n$ has $\dim(\mathbb{R}^n) = n$ vectors ([Theorem 7 in Subspaces and Rank](#), [Example 16 in Subspaces and Rank](#)). We show that $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$ are a basis of $\mathcal{C}(\mathbf{A})$, using [Theorem 4 in Subspaces and Rank](#) to apply $\mathbf{A}$ to linear combinations.

They are linearly independent. Suppose $\sum_{i=1}^{n-\nu} v_i \mathbf{A}\tilde{y}_i = \tilde{0}_m$. Then $\mathbf{A}(\sum_{i=1}^{n-\nu} v_i \tilde{y}_i) = \tilde{0}_m$ ([Theorem 4 in Subspaces and Rank](#)), so $\sum_{i=1}^{n-\nu} v_i \tilde{y}_i \in \mathcal{N}(\mathbf{A})$, and it equals $\sum_{j=1}^\nu u_j \tilde{x}_j$ for some numbers u_j , since the $\tilde{x}_j$ span $\mathcal{N}(\mathbf{A})$. Moving everything to one side,

$$\sum_{j=1}^\nu (-u_j) \tilde{x}_j + \sum_{i=1}^{n-\nu} v_i \tilde{y}_i = \tilde{0}_n,$$

and because the $\tilde{x}$'s and $\tilde{y}$'s together form a basis of $\mathbb{R}^n$, which is linearly independent, every coefficient is 0; in particular every $v_i = 0$.

They span $\mathcal{C}(\mathbf{A})$. Take any $\tilde{w} \in \mathcal{C}(\mathbf{A})$, so $\tilde{w} = \mathbf{A}\tilde{z}$ for some $\tilde{z} \in \mathbb{R}^n$ ([Definition 0.1](#)). Write $\tilde{z} = \sum_j a_j \tilde{x}_j + \sum_i b_i \tilde{y}_i$ in the basis of $\mathbb{R}^n$. Then

$$\begin{aligned} \tilde{w} &= \mathbf{A} \left(\sum_{j=1}^\nu a_j \tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \tilde{y}_i \right) \quad (\text{substitute } \tilde{z}) \\ &= \sum_{j=1}^\nu a_j \mathbf{A}\tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i \quad (\text{Theorem 4 in Subspaces and Rank}) \\ &= \sum_{j=1}^\nu a_j \tilde{0}_m + \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i \quad (\text{each } \tilde{x}_j \in \mathcal{N}(\mathbf{A})) \\ &= \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i, \quad (\text{drop the zero terms}) \end{aligned}$$

a linear combination of $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$. These vectors are in $\mathcal{C}(\mathbf{A})$, so their span is exactly $\mathcal{C}(\mathbf{A})$.

So $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$, and $\text{rank}(\mathbf{A}) = n - \nu$ by Theorem 0.5. □

Example 0.10 (Checking rank-nullity on the matrix in Example 0.1). The matrix $\mathbf{A}$ in Example 0.1 has $n = 3$ columns, rank 1 (Example 0.8) and nullity 2 (Example 0.9), and $1 + 2 = 3$. The 2×2 identity matrix has rank 2 and nullity 0 (Example 0.9), and $2 + 0 = 2$.

Theorem 0.7 (Multiplying on the right cannot increase the rank). *If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times k$, then*

$$\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A}).$$

i Proof

Proof. $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$. Any vector in $\mathcal{C}(\mathbf{AB})$ is $(\mathbf{AB})\tilde{x}$ for some $\tilde{x} \in \mathbb{R}^k$ (Definition 0.1), and

$$(\mathbf{AB})\tilde{x} = \mathbf{A}(\mathbf{B}\tilde{x}) \quad (\text{Theorem 5 in Matrices})$$

is $\mathbf{A}$ times the vector $\mathbf{B}\tilde{x} \in \mathbb{R}^n$, so it is in $\mathcal{C}(\mathbf{A})$.

Compare dimensions. Both column spaces are subspaces (Theorem 0.1), so

$$\begin{aligned} \text{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) \quad (\text{Theorem 0.5}) \\ &\leq \dim(\mathcal{C}(\mathbf{A})) \quad (\text{Theorem 10 in Subspaces and Rank, part 2}) \\ &= \text{rank}(\mathbf{A}). \quad (\text{Theorem 0.5}) \end{aligned}$$

□

Example 0.11 (A product can lose rank). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{AB} = \begin{bmatrix} 1 \cdot 0 + 0 \cdot 1 & 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 1, but $\mathbf{AB}$ has rank 0, so the inequality in Theorem 0.7 can be strict. With the 2×2 identity in place of $\mathbf{B}$, $\mathbf{AI} = \mathbf{A}$ keeps rank 1, and the inequality is an equality.

Theorem 0.8 (Fundamental theorem of ranks). *For any $m \times n$ matrix $\mathbf{A}$,*

$$\begin{aligned}
\text{rank}(\mathbf{A}) &= \text{rank}(\mathbf{A}^\top) \\
&= \text{rank}(\mathbf{A}^\top \mathbf{A}) \\
&= \text{rank}(\mathbf{A} \mathbf{A}^\top).
\end{aligned}$$

In particular, the largest number of linearly independent rows of $\mathbf{A}$ equals the largest number of linearly independent columns.

i Proof

Proof. $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$. Both matrices have n columns, and $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ (Theorem 0.4), so they have the same nullity (Definition 0.4). Then

$$\begin{aligned}
\text{rank}(\mathbf{A}^\top \mathbf{A}) &= n - \text{nullity}(\mathbf{A}^\top \mathbf{A}) \quad (\text{Theorem 0.6}) \\
&= n - \text{nullity}(\mathbf{A}) \quad (\text{equal null spaces}) \\
&= \text{rank}(\mathbf{A}). \quad (\text{Theorem 0.6})
\end{aligned}$$

$\text{rank}(\mathbf{A} \mathbf{A}^\top) = \text{rank}(\mathbf{A}^\top)$. Apply the previous step to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$.

Chain the inequalities.

$$\begin{aligned}
\text{rank}(\mathbf{A}) &= \text{rank}(\mathbf{A}^\top \mathbf{A}) \quad (\text{first step}) \\
&\leq \text{rank}(\mathbf{A}^\top) \quad (\text{Theorem 0.7, with } \mathbf{A}^\top \text{ on the left}) \\
&= \text{rank}(\mathbf{A} \mathbf{A}^\top) \quad (\text{second step}) \\
&\leq \text{rank}(\mathbf{A}). \quad (\text{Theorem 0.7, with } \mathbf{A} \text{ on the left})
\end{aligned}$$

The chain starts and ends at $\text{rank}(\mathbf{A})$, so every quantity in it equals $\text{rank}(\mathbf{A})$. The columns of $\mathbf{A}^\top$ are the rows of $\mathbf{A}$, which gives the statement about rows. $\square$

Example 0.12 (Four equal ranks). For $\mathbf{A}$ in Example 0.1, $\text{rank}(\mathbf{A}) = 1$ (Example 0.8).

Its transpose $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$ has second column 3 times its first, so its rank is 1. $\mathbf{A}^\top \mathbf{A}$

in Example 0.7 has every row a multiple of $(1, -2, -2)$, so its rank is 1 too, and

$$\begin{aligned}
\mathbf{A} \mathbf{A}^\top &= \begin{bmatrix} 1 \cdot 1 + (-2)(-2) + (-2)(-2) & 1 \cdot 3 + (-2)(-6) + (-2)(-6) \\ 3 \cdot 1 + (-6)(-2) + (-6)(-2) & 3 \cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} \\
&= \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix},
\end{aligned}$$

whose second column is 3 times its first, so its rank is 1.

Corollary 0.1 (The rank is at most the smaller dimension). *For any $m \times n$ matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$.*

i Proof

Proof. $\mathbf{A}$ has n columns, so at most n of them are linearly independent, and $\text{rank}(\mathbf{A}) \leq n$ ([Definition 2 in Subspaces and Rank](#)). Likewise $\mathbf{A}^\top$ has m columns, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) \leq m$ ([Theorem 0.8](#)). $\square$

Example 0.13 (A 2×3 matrix has rank at most 2). By [Corollary 0.1](#), every 2×3 matrix has rank at most $\min\{2, 3\} = 2$, even though it has 3 columns. The matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of [Remark 1 in Subspaces and Rank](#) reaches that bound: its first two columns $(1, 0)$ and $(0, 1)$ are linearly independent, so its rank is 2.

Definition 0.5 (Affine subspace). An **affine subspace** of $\mathbb{R}^p$ is a set of the form

$$\{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{S}\}$$

for some vector $\tilde{x}_0 \in \mathbb{R}^p$ and some subspace $\mathcal{S}$ of $\mathbb{R}^p$ ([Definition 4 in Subspaces and Rank](#)): the **image** of $\mathcal{S}$ under the translation by $\tilde{x}_0$ ([Definition 16 in Matrices](#)).

Example 0.14 (A line that misses the origin is an affine subspace). The line $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$ of [Example 4 in Subspaces and Rank](#) is not a subspace, but it is an affine subspace, with $\tilde{x}_0 = (0, 1)$ and $\mathcal{S} = \text{span}\{(1, 0)\}$ ([Definition 5 in Subspaces and Rank](#)):

$$\begin{aligned} (0, 1) + c(1, 0) &= (0 + c, 1 + 0) \quad (\text{Definition 6 in Matrices, Definition 6 in Vectors}) \\ &= (c, 1), \quad (\text{add}) \end{aligned}$$

and as c ranges over $\mathbb{R}$, $(c, 1)$ ranges over all of $\mathcal{T}$. Every subspace $\mathcal{S}$ is also an affine subspace, with $\tilde{x}_0 = \tilde{0}$.

Theorem 0.9 (A hyperplane is a translated subspace of dimension $p - 1$). *Let $\tilde{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w}^\top \tilde{x} + b = 0\}$ be the hyperplane with normal vector $\tilde{w}$ and offset b ([Definition 19 in Matrices](#)), and let $\mathcal{H}_0 = \{\tilde{x} \in \mathbb{R}^p : \tilde{w}^\top \tilde{x} = 0\}$ be the hyperplane with the same normal vector and offset 0.*

1. $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ ([Definition 4 in Subspaces and Rank](#)) of dimension $p - 1$ ([Definition 8 in Subspaces and Rank](#)).

2. $\mathcal{H}$ is not empty, and for any $\tilde{x}_0 \in \mathcal{H}$, $\mathcal{H} = \{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{H}_0\}$, so $\mathcal{H}$ is an affine subspace (Definition 0.5).
3. For any $\tilde{x}, \tilde{y} \in \mathcal{H}$, the normal vector is orthogonal to their difference: $\tilde{w} \perp (\tilde{x} - \tilde{y})$ (Definition 13 in Vectors).

i Proof

Proof. Here $\tilde{w}^\top$ is a $1 \times p$ matrix, and $\tilde{w}^\top \tilde{x} = \tilde{w} \cdot \tilde{x}$ (Example 6 in Matrices).

Part 1. $\mathcal{H}_0$ is the null space $\mathcal{N}(\tilde{w}^\top)$ (Definition 0.2), so it is a subspace (Theorem 0.2). The p columns of $\tilde{w}^\top$ are the numbers $w_1, \dots, w_p$. Some $w_j \neq 0$, because $\tilde{w} \neq \tilde{0}$, and that column on its own is linearly independent ($c w_j = 0$ forces $c = 0$), so $\text{rank}(\tilde{w}^\top) \geq 1$ (Definition 2 in Subspaces and Rank); and $\text{rank}(\tilde{w}^\top) \leq \min\{1, p\} = 1$ (Corollary 0.1). So

$$\begin{aligned} \dim(\mathcal{H}_0) &= \text{nullity}(\tilde{w}^\top) && \text{(Definition 0.4)} \\ &= p - \text{rank}(\tilde{w}^\top) && \text{(Theorem 0.6, for the } p \text{ columns of } \tilde{w}^\top) \\ &= p - 1. && \text{(the rank is 1)} \end{aligned}$$

Part 2. $\tilde{w} \cdot \tilde{w} = w_1^2 + \dots + w_p^2 > 0$, because some $w_j \neq 0$. So $\tilde{x}^* \stackrel{\text{def}}{=} -\frac{b}{\tilde{w} \cdot \tilde{w}} \tilde{w}$ is defined, and

$$\begin{aligned} \tilde{w}^\top \tilde{x}^* + b &= -\frac{b}{\tilde{w} \cdot \tilde{w}} (\tilde{w}^\top \tilde{w}) + b && \text{(homogeneity of } \tilde{x} \mapsto \tilde{w}^\top \tilde{x}, \text{ Theorem 9 in Matrices)} \\ &= -b + b && (\tilde{w}^\top \tilde{w} = \tilde{w} \cdot \tilde{w}) \\ &= 0, && \text{(arithmetic)} \end{aligned}$$

so $\tilde{x}^* \in \mathcal{H}$, and $\mathcal{H}$ is not empty. Now take any $\tilde{x}_0 \in \mathcal{H}$, so $\tilde{w}^\top \tilde{x}_0 = -b$. For any $\tilde{x} \in \mathbb{R}^p$,

$$\begin{aligned} \tilde{w}^\top (\tilde{x} - \tilde{x}_0) &= \tilde{w}^\top \tilde{x} - \tilde{w}^\top \tilde{x}_0 && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \tilde{w}^\top \tilde{x} - (-b) && (\tilde{x}_0 \in \mathcal{H}) \\ &= \tilde{w}^\top \tilde{x} + b, && \text{(arithmetic)} \end{aligned}$$

so $\tilde{x} \in \mathcal{H}$ exactly when $\tilde{x} - \tilde{x}_0 \in \mathcal{H}_0$, that is, exactly when $\tilde{x} = \tilde{x}_0 + \tilde{s}$ with $\tilde{s} = \tilde{x} - \tilde{x}_0 \in \mathcal{H}_0$. $\mathcal{H}_0$ is a subspace (part 1), so $\mathcal{H}$ has the form in Definition 0.5.

Part 3. For $\tilde{x}, \tilde{y} \in \mathcal{H}$, both $\tilde{w}^\top \tilde{x}$ and $\tilde{w}^\top \tilde{y}$ equal $-b$, so

$$\begin{aligned} \tilde{w} \cdot (\tilde{x} - \tilde{y}) &= \tilde{w}^\top \tilde{x} - \tilde{w}^\top \tilde{y} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= (-b) - (-b) && (\tilde{x}, \tilde{y} \in \mathcal{H}) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

□

Example 0.15 (The plane $x_1 + x_2 + x_3 = 1$ in $\mathbb{R}^3$). Take $\tilde{w} = (1, 1, 1)$ and $b = -1$, so $\mathcal{H} = \{\tilde{x} : x_1 + x_2 + x_3 - 1 = 0\}$, the plane that [Example 5 in Subspaces and Rank](#) showed is not a subspace. By part 1, $\mathcal{H}_0 = \{\tilde{x} : x_1 + x_2 + x_3 = 0\}$ is a subspace of dimension $3 - 1 = 2$.

- $\tilde{x}_0 = (1, 0, 0)$ is in $\mathcal{H}$, since $1 + 0 + 0 - 1 = 0$, and so is $\tilde{y} = (0, 1, 0)$.
- Their difference $\tilde{y} - \tilde{x}_0 = (-1, 1, 0)$ has entries summing to $-1 + 1 + 0 = 0$, so it is in $\mathcal{H}_0$, and $\tilde{y} = \tilde{x}_0 + (-1, 1, 0)$, as part 2 says.

•

$$\begin{aligned}\tilde{w} \cdot (-1, 1, 0) &= -1 + 1 + 0 \\ &= 0,\end{aligned}$$

as part 3 says.

- The point built in the proof is

$$\begin{aligned}\tilde{x}^* &= -\frac{-1}{3} (1, 1, 1) \\ &= \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right),\end{aligned}$$

$$\text{and } \frac{1}{3} + \frac{1}{3} + \frac{1}{3} - 1 = 0.$$

Theorem 0.10 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$). *Let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b = 0\}$ be a hyperplane ([Definition 19 in Matrices](#)).*

1. $\mathcal{H}$ contains the origin $\tilde{0}$ exactly when $b = 0$.
2. If $p = 2$, then $\mathcal{H}$ is a line: $\mathcal{H} = \{\tilde{x}_0 + t\tilde{v} : t \in \mathbb{R}\}$ for some $\tilde{x}_0 \in \mathbb{R}^2$ and some nonzero $\tilde{v} \in \mathbb{R}^2$.
3. If $p = 3$, then $\mathcal{H}$ is a plane: $\mathcal{H} = \{\tilde{x}_0 + s\tilde{v}_1 + t\tilde{v}_2 : s, t \in \mathbb{R}\}$ for some $\tilde{x}_0 \in \mathbb{R}^3$ and some linearly independent $\tilde{v}_1, \tilde{v}_2 \in \mathbb{R}^3$.

i Proof

Proof. **Part 1.** Since $\tilde{w} \cdot \tilde{0} = 0$,

$$\begin{aligned}\tilde{w} \cdot \tilde{0} + b &= 0 + b \quad (\text{the dot product with } \tilde{0} \text{ is } 0) \\ &= b, \quad (\text{Theorem ??})\end{aligned}$$

so $\tilde{0} \in \mathcal{H}$ exactly when $b = 0$.

Parts 2 and 3. By [Theorem 0.9](#), $\mathcal{H} = \{\tilde{x}_0 + \tilde{s} : \tilde{s} \in \mathcal{H}_0\}$ for any $\tilde{x}_0 \in \mathcal{H}$, where $\mathcal{H}_0$ is

a subspace of $\mathbb{R}^p$ of dimension $p - 1$.

For $p = 2$, $\mathcal{H}_0$ has dimension 1, so it has a basis ([Definition 6 in Subspaces and Rank](#)) with one vector $\tilde{v}$. That vector is nonzero, and $\mathcal{H}_0 = \{t\tilde{v} : t \in \mathbb{R}\}$.

For $p = 3$, $\mathcal{H}_0$ has dimension 2, so it has a basis with two vectors $\tilde{v}_1$ and $\tilde{v}_2$. They are linearly independent, and $\mathcal{H}_0 = \{s\tilde{v}_1 + t\tilde{v}_2 : s, t \in \mathbb{R}\}$. $\square$

Example 0.16 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points). In $\mathbb{R}^2$, take $\tilde{w} = (1, 1)$ and $b = -1$, the line of [Example 19 in Matrices](#). The point $\tilde{x}_0 = -\frac{b}{\tilde{w} \cdot \tilde{w}} \tilde{w}$ is on it, and the vector $\tilde{v} = (-w_2, w_1)$ is nonzero with $\tilde{w} \cdot \tilde{v} = 0$, so it spans the subspace $\mathcal{H}_0$ of Theorem 0.9. In $\mathbb{R}^3$, take $\tilde{w} = (1, 1, 1)$ and $b = -1$, the plane of [Example 0.15](#), with the independent vectors $\tilde{v}_1 = (1, -1, 0)$ and $\tilde{v}_2 = (0, 1, -1)$ in $\mathcal{H}_0$. The code evaluates $\tilde{w} \cdot \tilde{x} + b$ at the origin and at points $\tilde{x}_0 + t\tilde{v}$ and $\tilde{x}_0 + s\tilde{v}_1 + t\tilde{v}_2$.

```
level <- function(w, b, x) sum(w * x) + b

w2 <- c(1, 1)
b2 <- -1
point_line <- -b2 / sum(w2 * w2) * w2
v <- c(-w2[2], w2[1])
line_levels <- sapply(-2:2, function(t) level(w2, b2, point_line + t * v))

w3 <- c(1, 1, 1)
b3 <- -1
point_plane <- -b3 / sum(w3 * w3) * w3
v1 <- c(1, -1, 0)
v2 <- c(0, 1, -1)
grid <- expand.grid(s = -2:2, t = -2:2)
plane_levels <- mapply(
  function(s, t) level(w3, b3, point_plane + s * v1 + t * v2),
  grid$s, grid$t
)

c(
  origin_in_R2 = level(w2, b2, c(0, 0)),
  origin_in_R3 = level(w3, b3, c(0, 0, 0)),
  origin_with_b0 = level(w3, 0, c(0, 0, 0))
)
#>   origin_in_R2   origin_in_R3 origin_with_b0
#>         -1         -1         0
```

At the origin the left side equals the offset, so the origin is on the line only if b is 0: it

gives -1 for the line and -1 for the plane, and 0 once the offset is changed to 0. All 5 points $\tilde{x}_0 + t\tilde{v}$ on the line and all 25 points $\tilde{x}_0 + s\tilde{v}_1 + t\tilde{v}_2$ on the plane satisfy the equation up to rounding error: 30 of the 30 points have $|\tilde{w} \cdot \tilde{x} + b| < 10^{-10}$.

Definition 0.6 (Half-space). Let $\mathcal{H} = \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b = 0\}$ be a hyperplane (Definition 19 in Matrices). The two **open half-spaces** of $\mathcal{H}$ are

$$\mathcal{H}^+ \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b > 0\} \quad \text{and} \quad \mathcal{H}^- \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^p : \tilde{w} \cdot \tilde{x} + b < 0\}. \quad (1)$$

The **closed half-spaces** also include $\mathcal{H}$ itself: $\mathcal{H}^+ \cup \mathcal{H}$ and $\mathcal{H}^- \cup \mathcal{H}$. Every point of $\mathbb{R}^p$ is in exactly one of $\mathcal{H}^+$, $\mathcal{H}$ and $\mathcal{H}^-$, because the number $\tilde{w} \cdot \tilde{x} + b$ is positive, zero or negative. The vector $\tilde{w}$ points into $\mathcal{H}^+$: the number $\tilde{w} \cdot \tilde{x} + b$ goes up when $\tilde{x}$ moves in the direction $\tilde{w}$.

Remark. Replacing $(\tilde{w}, b)$ by $(-\tilde{w}, -b)$ describes the same hyperplane but swaps the names $\mathcal{H}^+$ and $\mathcal{H}^-$. The sign of $\tilde{w} \cdot \tilde{x} + b$ tells you which side of the hyperplane $\tilde{x}$ is on. A linear classifier uses this sign to assign one of two labels.

Example 0.17 (Half-spaces of a line in the plane). Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\tilde{w} = (1, 2)$ and $b = -2$ (Definition 19 in Matrices), so $\mathcal{H} = \{\tilde{x} : x_1 + 2x_2 - 2 = 0\}$. Evaluate $\tilde{w} \cdot \tilde{x} + b$ at three points (Definition 0.6):

- At $(0, 0)$ it equals $0 + 0 - 2 = -2 < 0$, so $(0, 0) \in \mathcal{H}^-$.
- At $(2, 0)$ it equals $2 + 0 - 2 = 0$, so $(2, 0) \in \mathcal{H}$.
- At $(2, 1)$ it equals $2 + 2 - 2 = 2 > 0$, so $(2, 1) \in \mathcal{H}^+$.

Moving from $(2, 0)$ in the direction $\tilde{w} = (1, 2)$ gives $(3, 2)$, where the value is $3 + 4 - 2 = 5 > 0$. So the direction $\tilde{w}$ leads into $\mathcal{H}^+$.

Theorem 0.11 (Differences of points in a hyperplane and in a half-space). Let $\tilde{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, and let $\mathcal{H}$, $\mathcal{H}^+$ be the hyperplane and open half-space of Definition 0.6. For a set $S \subseteq \mathbb{R}^p$, let $D(S) \stackrel{\text{def}}{=} \{\tilde{y} - \tilde{x} : \tilde{x}, \tilde{y} \in S\}$ be the set of differences of its points.

1. $D(\mathcal{H}) = \{\tilde{s} : \tilde{w} \cdot \tilde{s} = 0\}$, a subspace of dimension $p - 1$.
2. $D(\mathcal{H}^+) = \mathbb{R}^p$, a subspace of dimension p .

The same holds for $\mathcal{H}^-$.

i Proof

Proof. Part 1. Let $\tilde{x}, \tilde{y} \in \mathcal{H}$. By Theorem 0.9 part 3, $\tilde{w} \cdot (\tilde{y} - \tilde{x}) = 0$, so $\tilde{y} - \tilde{x} \in \mathcal{H}_0 = \{\tilde{s} : \tilde{w} \cdot \tilde{s} = 0\}$. Conversely, fix any $\tilde{x}_0 \in \mathcal{H}$ (it exists by Theorem 0.9 part 2). For any $\tilde{s} \in \mathcal{H}_0$, the point $\tilde{x}_0 + \tilde{s}$ is in $\mathcal{H}$ by the same part, and $(\tilde{x}_0 + \tilde{s}) - \tilde{x}_0 = \tilde{s}$. So $D(\mathcal{H}) = \mathcal{H}_0$, which has dimension $p - 1$ by Theorem 0.9 part 1.

Part 2. Every difference is in $\mathbb{R}^p$, so $D(\mathcal{H}^+) \subseteq \mathbb{R}^p$. For the other direction, fix any $\tilde{x}_0 \in \mathcal{H}$ and any $\tilde{v} \in \mathbb{R}^p$. For a number $t > 0$ to be chosen, let $\tilde{x}_t \stackrel{\text{def}}{=} \tilde{x}_0 + t\tilde{w}$. Then

$$\begin{aligned} \tilde{w} \cdot \tilde{x}_t + b &= (\tilde{w} \cdot \tilde{x}_0 + b) + t\tilde{w} \cdot \tilde{w} \quad (\text{linearity of } \tilde{x} \mapsto \tilde{w} \cdot \tilde{x}) \\ &= t\tilde{w} \cdot \tilde{w}, \quad (\tilde{x}_0 \in \mathcal{H}) \end{aligned}$$

which is positive, so $\tilde{x}_t \in \mathcal{H}^+$. Next,

$$\begin{aligned} \tilde{w} \cdot (\tilde{x}_t + \tilde{v}) + b &= (\tilde{w} \cdot \tilde{x}_t + b) + \tilde{w} \cdot \tilde{v} \quad (\text{linearity of } \tilde{x} \mapsto \tilde{w} \cdot \tilde{x}) \\ &= t\tilde{w} \cdot \tilde{w} + \tilde{w} \cdot \tilde{v}. \quad (\text{the display above}) \end{aligned}$$

Because $\tilde{w} \cdot \tilde{w} > 0$, this is positive once $t > -\tilde{w} \cdot \tilde{v} / \tilde{w} \cdot \tilde{w}$. Choose such a $t > 0$. Then $\tilde{x}_t$ and $\tilde{x}_t + \tilde{v}$ are both in $\mathcal{H}^+$, and their difference is $\tilde{v}$. So $\tilde{v} \in D(\mathcal{H}^+)$. The whole space $\mathbb{R}^p$ is a subspace of dimension p (Definition 8 in Subspaces and Rank).

The proof for $\mathcal{H}^-$ is the same with $-\tilde{w}$ and $-b$ in place of $\tilde{w}$ and b . □

Example 0.18 (Two iris species on either side of a line). The `iris` data set in R has the petal length and petal width, in centimeters, of 150 irises. Take the 100 flowers of the species `versicolor` and `virginica`, and let $\tilde{x} = (x_1, x_2)$ be the petal length and petal width of one flower. A logistic regression of the species on $\tilde{x}$ gives a coefficient vector $\tilde{w}$ and an intercept b . The line $\tilde{w} \cdot \tilde{x} + b = 0$ is a hyperplane in $\mathbb{R}^2$ (Definition 19 in Matrices), and the two species mostly fall in its two open half-spaces (Definition 0.6). Figure 1 shows the line and the flowers.

```

two_species <- dplyr::filter(iris, Species != "setosa") |> droplevels()
fit <- glm(
  Species ~ Petal.Length + Petal.Width,
  family = binomial,
  data = two_species
)
w <- coef(fit)[c("Petal.Length", "Petal.Width")]
b <- coef(fit)[["(Intercept)"]]
two_species$score <- as.vector(
  as.matrix(two_species[c("Petal.Length", "Petal.Width")]) %*% w + b
)
two_species$side <- ifelse(two_species$score > 0, "positive", "negative")
ggplot2::ggplot(two_species) +
  ggplot2::aes(x = Petal.Length, y = Petal.Width, colour = Species) +
  ggplot2::geom_point() +
  ggplot2::geom_abline(intercept = -b / w[[2]], slope = -w[[1]] / w[[2]]) +
  ggplot2::labs(x = "Petal length (cm)", y = "Petal width (cm)")

```

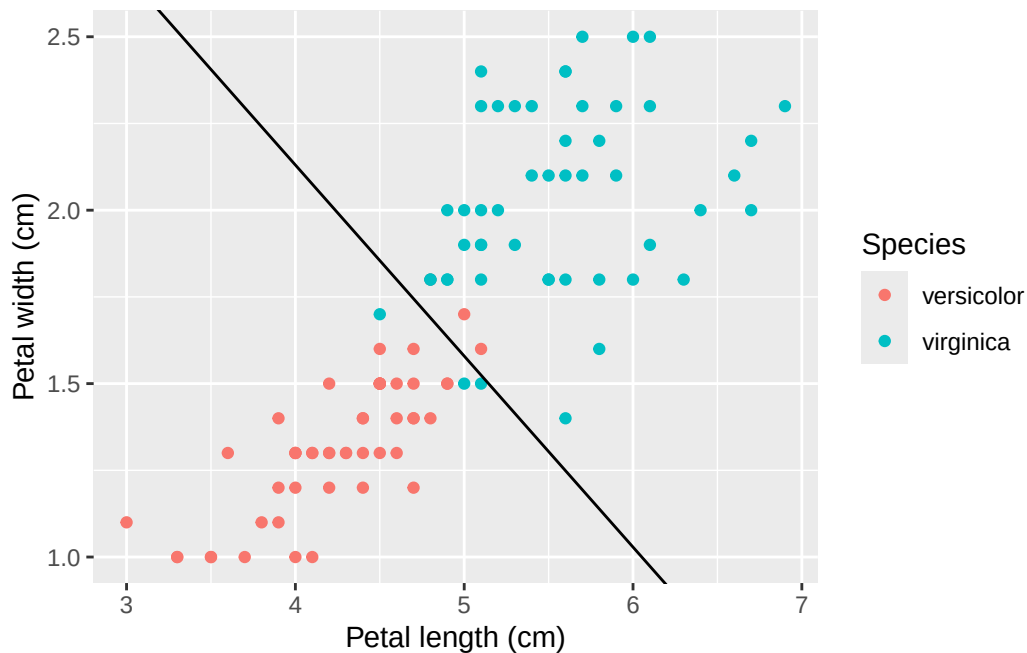


Figure 1: Petal length and width of versicolor and virginica irises, with the line where a logistic regression is indifferent between the two species

```
table(two_species$Species, two_species$side)
#>
#>           negative positive
#> versicolor      47        3
#> virginica       3        47
```

The vector $\tilde{w}$ points toward the virginica side, so the positive half-space $\mathcal{H}^+$ holds 50 of the 100 flowers, and 47 of them are virginica. The line is a set of dimension $p - 1 = 1$, and each side is a region of the plane of dimension $p = 2$ (Theorem 0.11).

Exercise 0.1 (Which side of the plane?). Let $\mathcal{H}$ be the plane in $\mathbb{R}^3$ with $\tilde{w} = (1, 1, 1)$ and $b = -1$ (Example 0.15).

1. For each point, say whether it is in $\mathcal{H}^+$, $\mathcal{H}$ or $\mathcal{H}^-$ (Definition 0.6): $(0, 0, 0)$, $(1, 0, 0)$ and $(1, 1, 1)$.
2. Write the plane and its half-spaces using $-\tilde{w}$ and $-b$. Which points from part 1 change sides?
3. State the dimension of $D(\mathcal{H})$ and of $D(\mathcal{H}^+)$ (Theorem 0.11).

Solution

Solution 0.1.

1. Compute $\tilde{w} \cdot \tilde{x} + b = x_1 + x_2 + x_3 - 1$ at each point.
 - At $(0, 0, 0)$ it equals $0 + 0 + 0 - 1 = -1 < 0$, so the point is in $\mathcal{H}^-$.
 - At $(1, 0, 0)$ it equals $1 + 0 + 0 - 1 = 0$, so the point is in $\mathcal{H}$.
 - At $(1, 1, 1)$ it equals $1 + 1 + 1 - 1 = 2 > 0$, so the point is in $\mathcal{H}^+$.
2. With $-\tilde{w} = (-1, -1, -1)$ and $-b = 1$, the equation is $-x_1 - x_2 - x_3 + 1 = 0$. Multiplying both sides by -1 gives the original equation, so it is the same plane. The new expression is the negative of the old one at every point, so the new positive half-space is the old $\mathcal{H}^-$ and the new negative half-space is the old $\mathcal{H}^+$. The origin $(0, 0, 0)$ is now in the positive half-space, and $(1, 1, 1)$ is now in the negative half-space. The point $(1, 0, 0)$ stays on the plane. So the origin and $(1, 1, 1)$ both change names, and the point on the plane does not.
3. Here $p = 3$. $D(\mathcal{H})$ is the subspace $\{\tilde{s} : s_1 + s_2 + s_3 = 0\}$, which has dimension $p - 1 = 2$. $D(\mathcal{H}^+) = \mathbb{R}^3$ has dimension $p = 3$.

Definition 0.7 (Rank factorization). Let $\mathbf{A}$ be an $m \times n$ matrix with $\text{rank}(\mathbf{A}) = r \geq 1$. A **rank factorization** of $\mathbf{A}$ is a product

$$\underset{m \times n}{\mathbf{A}} = \underset{m \times r}{\mathbf{C}} \underset{r \times n}{\mathbf{R}}.$$

Example 0.19 (A rank factorization of a rank-one matrix). The matrix $\mathbf{A}$ in Example 0.1 has rank 1 (Example 0.8), and

$$\begin{aligned} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \end{bmatrix} &= \begin{bmatrix} 1 \cdot 1 & 1 \cdot (-2) & 1 \cdot (-2) \\ 3 \cdot 1 & 3 \cdot (-2) & 3 \cdot (-2) \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}, \end{aligned}$$

so $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ (1×3) are a rank factorization. It is not unique: $\mathbf{C} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} \frac{1}{2} & -1 & -1 \end{bmatrix}$ give the same product. A product $\mathbf{C}\mathbf{R}$ with $\mathbf{C}$ of size 2×2 is not a rank factorization of this $\mathbf{A}$, because the inner dimension must equal $\text{rank}(\mathbf{A}) = 1$.

Theorem 0.12 (Every nonzero matrix has a rank factorization). *Every $m \times n$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = r \geq 1$ has a rank factorization (Definition 0.7). One is given by taking the columns of $\mathbf{C}$ to be any basis of $\mathcal{C}(\mathbf{A})$.*

Proof

Proof. $\mathcal{C}(\mathbf{A})$ has dimension r (Theorem 0.5), so it has a basis $\tilde{c}_1, \dots, \tilde{c}_r$ (Theorem 9 in Subspaces and Rank); let $\mathbf{C}$ be the $m \times r$ matrix with these columns. Each column $\tilde{a}_j$ of $\mathbf{A}$ is in $\mathcal{C}(\mathbf{A})$, so $\tilde{a}_j = r_{1j}\tilde{c}_1 + \dots + r_{rj}\tilde{c}_r$ for some numbers r_{ij} , because the basis spans $\mathcal{C}(\mathbf{A})$. Let $\mathbf{R}$ be the $r \times n$ matrix with entries r_{ij} , whose column j is $\tilde{r}_j = (r_{1j}, \dots, r_{rj})$. Then column j of $\mathbf{C}\mathbf{R}$ is

$$\begin{aligned} \mathbf{C}\tilde{r}_j &= r_{1j}\tilde{c}_1 + \dots + r_{rj}\tilde{c}_r \quad (\text{Theorem 3 in Subspaces and Rank}) \\ &= \tilde{a}_j, \quad (\text{choice of the } r_{ij}) \end{aligned}$$

so $\mathbf{C}\mathbf{R} = \mathbf{A}$. □

Example 0.20 (Building a rank factorization from a basis). For $\mathbf{A}$ in Example 0.1, $(1, 3)$ spans $\mathcal{C}(\mathbf{A})$ (Example 0.2), and it is linearly independent because it is nonzero ($c(1, 3) = \tilde{0}_2$ forces $c = 0$), so it is a basis of $\mathcal{C}(\mathbf{A})$. The columns of $\mathbf{A}$ are $(1, 3) = 1 \cdot (1, 3)$,

$(-2, -6) = -2 \cdot (1, 3)$ and $(-2, -6) = -2 \cdot (1, 3)$, so the construction in Theorem 0.12 gives $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$, the factorization in Example 0.19.

Zhou, Hua. 2024. *Rank and Nullity*. Lecture notes for Biostat 216, Mathematical Methods for Biostatistics, University of California, Los Angeles. <https://ucla-biostat-216.github.io/2024fall/slides/05-rank/05-rank.html>.